



**Local AI for
Genealogists**
Model Guide — Which
AI Model Should I Use?
Seattle Genealogical
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Model Guide: Which Model Should I Use?

A “model” is a large file that contains a trained AI brain. Different models are optimized for different tasks. You download models inside LM Studio and switch between them as needed.

You do not need all of these. Start with one.

Quick Selection Guide

My Goal	Best Model	Minimum RAM
Ask genealogy questions about text I paste	Qwen3 8B	8 GB
Analyze an image of a handwritten record	Qwen3 VL 8B	16 GB
Best quality image analysis	Qwen3 VL 32B	32 GB
	Nomic Embed + Qwen3	16 GB

My Goal	Best Model	Minimum RAM
Search across a folder of documents (August)		
Compare with a different AI brand	Gemma 3	16 GB

Recommended starting point: Qwen3 VL 8B (works on most Mac computers with 16 GB memory; can analyze both text and images of records)

Detailed Model Descriptions

Qwen3 8B

What it is: A text-only language model from Alibaba’s Qwen team. The “8B” means 8 billion parameters — a measure of the model’s complexity.

What it does well: - Answering genealogy questions when you paste text - Summarizing research notes and transcriptions - Writing research log entries and biographical narratives - Planning research strategies - Formatting Evidence Explained citations from information you provide

What it cannot do: Read images. You must paste the text yourself.

Size: ~5 GB download **Speed:** Very fast — responses in 5–15 seconds on most machines **RAM needed:** 8 GB minimum, 16 GB comfortable

Best for: Text-based work, writing, planning, summarizing

Qwen3 VL 8B

What it is: The same Qwen3 model with “VL” added — Vision Language. It can see and read images of documents.

What it does well: - Reading photographs of handwritten census records - Transcribing marriage certificates and death records - Analyzing land patents and court records - Everything Qwen3 8B does, plus images

Size: ~8 GB download **Speed:** Fast — 20–40 seconds for image analysis on M1/M2 Macs **RAM needed:** 16 GB recommended

Best for: Most genealogy research that involves scanned documents. This is the recommended starting model for most researchers.

Qwen3 VL 32B

What it is: A larger version of the vision language model. The “32B” means 32 billion parameters — a much larger brain.

What it does well: - Superior accuracy on difficult handwriting - Better at reading faded, damaged, or unusual scripts - More reliable on documents with complex layouts - Higher quality narrative writing

What the size difference means in practice: The VL 32B produces noticeably better results on challenging 19th-century handwriting compared to VL 8B. The trade-off is that it is slower and requires more memory.

Size: ~20 GB download **Speed:** Moderate — 45–90 seconds per image on M2/M3 Macs with 24+ GB **RAM needed:** 32 GB recommended; possible on 24 GB but slower

Best for: Difficult handwriting, complex records, maximum accuracy. Used in tonight’s live demonstration.

Gemma 3 (Google)

What it is: Google’s open-source language model. Available in several sizes (4B, 12B, 27B). A good alternative to Qwen3.

What it does well: - Similar capabilities to Qwen3 8B - Different “voice” — useful for comparing outputs - Good writing quality for biographical narratives

What it cannot do: The basic Gemma models do not have vision capability. (Check LM Studio’s model page for the latest Gemma VL variant if available.)

Size: Varies by version (4B: ~4 GB, 12B: ~8 GB, 27B: ~18 GB) **Speed:** Fast to moderate depending on size **RAM needed:** 16 GB for 12B version

Best for: Comparison with Qwen3 outputs; narrative writing; researchers who prefer Google’s model lineage

Nomic Embed Text v1.5

What it is: This is NOT a chat model. It is an “embedding model” — a specialized AI that converts documents into searchable vectors.

What it does: Allows EverythingLLM (covered in August) to search across your genealogy document collection. It reads your documents and creates a searchable index.

What it cannot do: Answer questions or generate text. You cannot chat with it directly.

Size: ~1 GB download **RAM needed:** Minimal

Best for: Document search across large collections (August workshop)

How Models Are Named

Understanding model names helps when browsing the LM Studio model library.

Part of Name	What It Means
Qwen3	Made by Alibaba; third generation
VL	Vision Language — can read images
8B, 14B, 32B	Number of parameters (larger = more capable)
Gemma	Made by Google
Nomic Embed	Made for search, not chat
GGUF	File format for Windows/CPU (works without GPU)
MLX	File format optimized for Apple Silicon Mac

When downloading inside LM Studio: If you see multiple versions of the same model, choose MLX format for Mac, GGUF for Windows.

Model Sizes and Download Times

Model	Size	Download Time (typical)
Qwen3 8B	~5 GB	10–20 minutes
Qwen3 VL 8B	~8 GB	15–30 minutes
Qwen3 VL 32B	~20 GB	45–90 minutes
Nomic Embed	~1 GB	3–5 minutes

Download once; use forever. No re-downloading unless you choose to update.

Models NOT Recommended for Beginners

- **DeepSeek models:** Excellent reasoning, but have raised privacy concerns about their servers (less relevant for local use, but worth knowing)
 - **Llama (Meta):** Good models, but Qwen3 VL is better for document images
 - **Mistral:** Fast and capable, but no vision capability in most versions
 - **70B+ models:** Generally require 64+ GB of RAM — too large for most laptops
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